# Art Pipeline

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# 1 Introduction

This document describes the art pipeline for authoring and previewing an art asset.

A word about the examples in this/these documents.

We used the modeling package 3D Studio max to create Rail Simulator, so this product appears in the examples we have provided. You can, of course, use other packages such as Blender or 3D Canvas. The package we used for authoring textures is Adobe Photoshop, again you may use one of the many other packages available. Regardless of the packages you use, the document should be used as a general guide to the processes to follow to build assets for Rail Simulator

Important note: For your chosen package, you may need additional plug-ins to export the assets into Rail Simulator. We are in the process of finalising these tools and so plug-ins for some packages may not be available immediately.

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# 2 Basic Pipeline and Plug-ins

## 2.1 Basic Pipeline

### 2.1.1 Textures

Texture assets are authored as layered Photoshop PSD files for flexibility and backup purposes. These textures must then be exported as ACE files ready to be converted by the game tools.

### 2.1.2 Geometry

The geometry assets will be exported from Max as IGS (intermediate geometry) files.

## 2.2 ACE Format

Rail Simulator supports Kuju's own proprietary texture format - the ACE format.

### 2.2.1 Texture sizes

The ACE format only supports texture sizes which are 'the order of 2' and can support non-square textures e.g. 256x1024, 512x512, 64x128, 1024x1024, 128x512 etc

### 2.2.2 Colour depth

The ACE format supports 8-bit, 24-bit or 32-bit
Any texture that requires transparency should be saved as a full 32-bit image, with the alpha channel containing only black or white (not grey scale). If the image contains grey scale alpha, the DX compression will simply quantise this down to 1-bit, and the resultant alpha channel may be undesirable. With transparency, remember to set the TRANS flag in the shader.

## 2.3 MIPs

The ACE format supports textures with or without MIPs.

Generally textures require MIPs; however in some cases (for example UI, sky domes, etc) it's a waste of texture space to stores MIPs for a texture that will never be viewed at a distance from the camera.

The ACE plug-in has an extra MIP dialogue box which is used on opening or saving a texture. There are two options 'make MIPs' and 'show MIPs'

### 2.3.1 MAKE MIPs (auto-generates MIPs when saving)

This is the suggested option to use if the user simply requires a texture with MIPs.

If this option is used on saving, the plug-in will take a texture and generate the MIPS for that texture regardless of whether it had MIPs in the first place.

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NOTE: If the texture had any hand-edited MIPs, then those MIPs will be overwritten with auto-generated ones if this option is used on saving. If the user wishes hand–edited MIPs to be preserved then they must save out the texture using the option detailed below.

#### 2.3.2 SHOW MIPS (saves the texture without auto-generating the MIPs)

This option has 2 uses:

With this option enabled, the user can open a file and view (and edit) the MIPs of a texture. This is the suggested option to use if the user wishes to preserve any hand-edited MIPs. The user can then make the edits and save ensuring that the 'show MIPs' option is still enabled.

This is also the option to use if the user wishes to save out the texture with no MIPs. Initially open the texture (with the 'make MIPs' option enabled). The user can crop off the MIPs in Photoshop (simply crop off the bottom half of the image) and save the file with the show MIPs option enabled.

### 2.4 Kuju material

The Kuju material plug-in in Max allows materials and textures to be applied to assets ready for exporting.

See the shaders doc for more information.

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# 3 The Asset Editor

## 3.1 Summary

The Asset Editor allows the user to preview an asset and export it into the game if desired.

## 3.2 Controls

There are two modes in which to view the previewed asset; 'Free Camera Mode' and 'Target Camera Mode'.

### Free camera mode

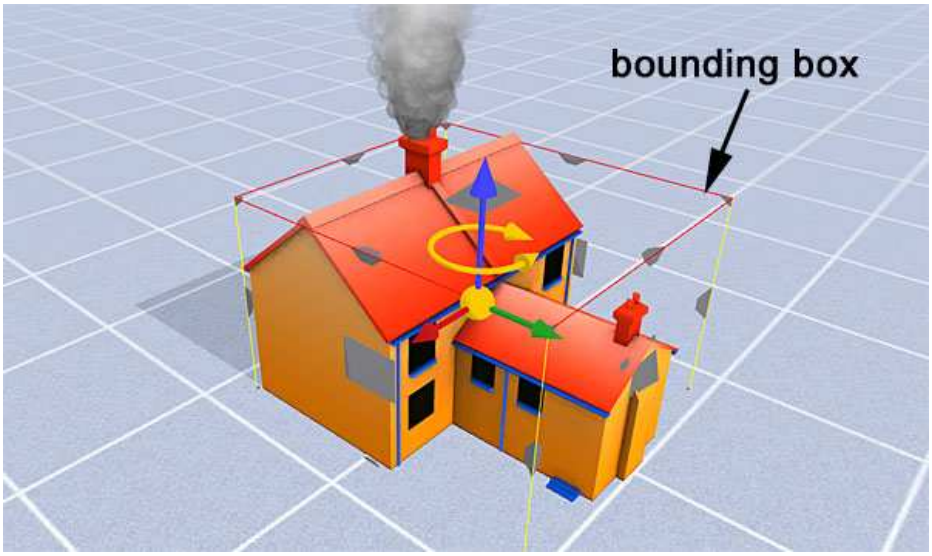
| | | |
|---|---|---|
| Keyboard cursor keys | : | Camera position |
| Right mouse button +moving | : | Camera target or look at |
| Left mouse button | : | Selects object and object gizmo |

### Target camera mode

| | | |
|---|---|---|
| Keyboard cursor keys | : | Camera orbits around the asset |

## 3.3 Bounding Boxes

Assets that require solid collision must have a bounding box otherwise they will simply be ignored in collision calculations. The bounding box for an asset should be created to give a simple representation of the shape of the asset, so that the physics calculations are kept to minimum.




The bounding boxes are created in the Asset Editor and are stored in the blueprint of the asset.

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Bounding boxes serve two main purposes:

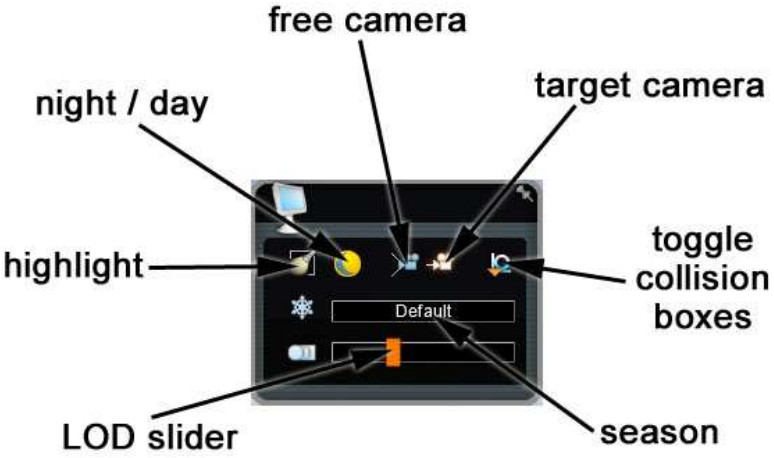
- Collisions in train derailments
- Passenger avoidance on platforms

Once the train is derailed, the bounding boxes are used to calculate collisions between rolling stock and scenery objects.

On platforms, the animated passengers will avoid walking through assets with bounding boxes.

### 3.4 Extra Features

There are a number of buttons in the Asset Editor which will help to debug an asset. These are as follows:

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# 4 Exporting and Viewing Art

## 4.1 Basics

In Max, the Max file is exported to an Intermediate Geometry Shape file (IGS)

**Max file -> IGS file**

In the Asset Editor, the game ready data is compiled from the IGS file blueprint

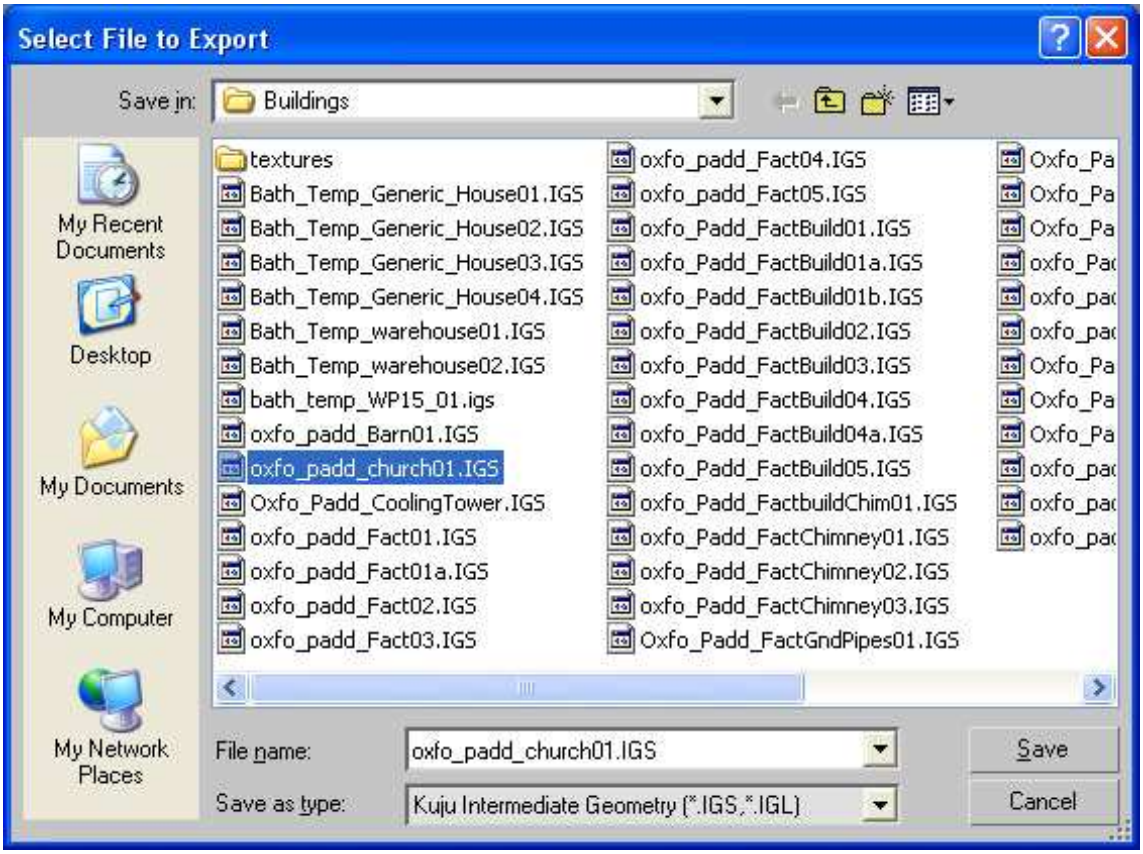
**IGS file -> GEOPCDX file**

**ACE file -> TGPCDX file**

The GEOPCDX (geometry) and TGPCDX (textures) are 'game ready' file formats.

## 4.2 Export IGS from Max

With the asset in Max, choose FILE EXPORT and you will see the screen:

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Choose the correct filename (and the IGS file type) for your asset.

- The exported IGS filename must be the same as the MAX filename and exported into the same location as the max file

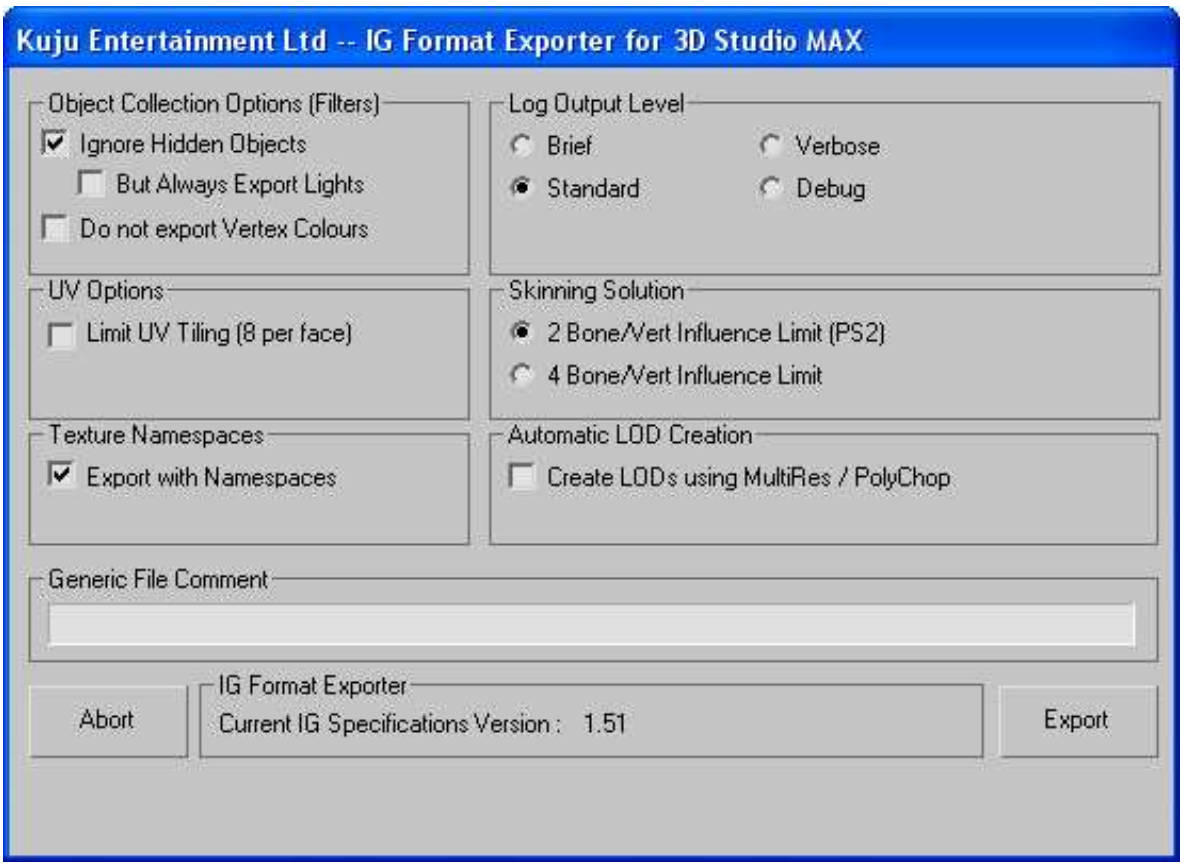
For example, consider the asset called oxfo_padd_church01

**C:\dev\railsim\Source\Scenery\Buildings\oxfo_padd_church01.MAX**

exports to:

**C:\dev\railsim\Source\Scenery\Buildings\oxfo_padd_church01.IGS**

Once you've named the IGS file, use the following options on export




Click on **EXPORT**

Ensure that your exported "Level of Detail" nodes are in numerical order in the export dialogue window (see below)

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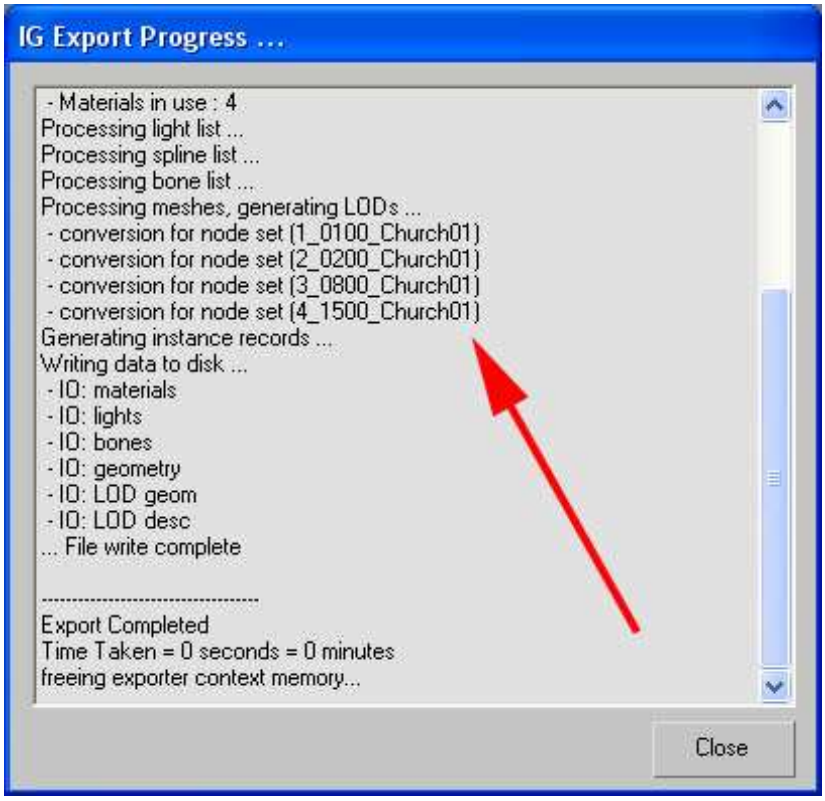


If they appear in the wrong order, then this may cause export failures. The LODs must be created in max in the correct numeric order to get the LODs to export in the correct order. Even though they are named correctly (1_..., 2_..., 3_... etc), they may appear in the wrong order if they were created in the wrong order in Max.

## 4.3 Previewing an IGS in the Asset Editor

### 4.3.1 Start up the Asset Editor

An environment appears with a Blueprint Editor Browser window on the left. In this browser window, the source art is represented in a directory structure

### 4.3.2 Previewing an asset

The Asset Editor requires an asset blueprint to preview an asset.

You must either create a new blueprint for your asset (if one doesn't already exist) or use a previously created blueprint.
Creating a blueprint
Browse to your IGS file
Right Click on the IGS file to get this window, and select "New Shape Blueprint"

For a scenery object, such as a building, then select Scenery Blueprint from the list;

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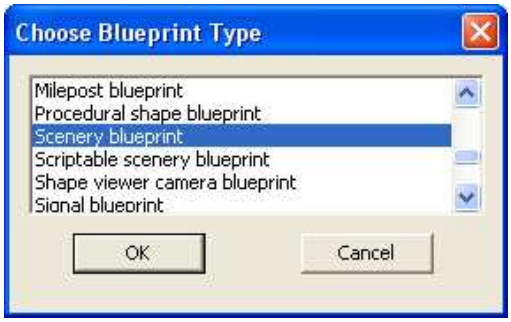
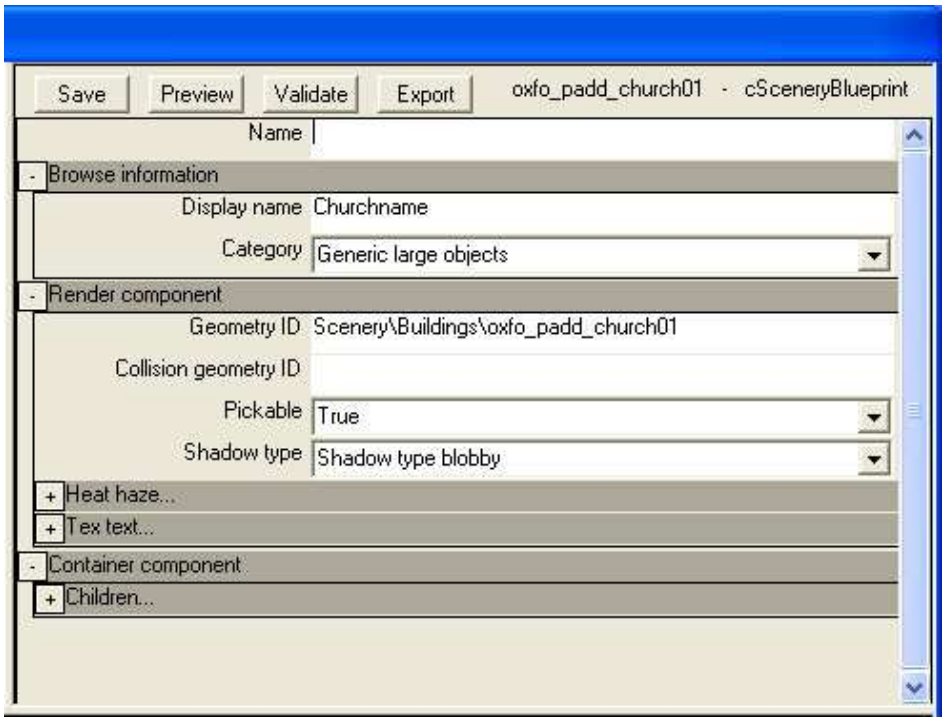



Give the asset a "Display name". For example "Churchname"
Set the correct category type. For example "Generic large objects"

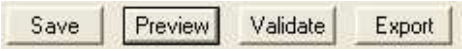



This blueprint can now be used to preview the asset in the Asset Editor.

**4.3.3 Viewing the asset**

If you have just created the blueprint, a simple click on the Preview button is required to view the asset.

If you are viewing a shape that already has a blueprint, then initially you must double-click on the blueprint in the browser to open the blueprint. Then a simple click on the Preview button is required to view the asset.

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In order that the Asset Editor can display the asset, the Asset Editor will create a temporary version of the compiled data (usually in a shape-viewer folder) and use that to display.

### 4.4 Exporting the Asset

Clicking on Export will copy the compiled files (GEOPCDX and TGPCDX in the temporary folder) to the appropriate place for the compiled data in your local game folder, so they can be used locally by the game.

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